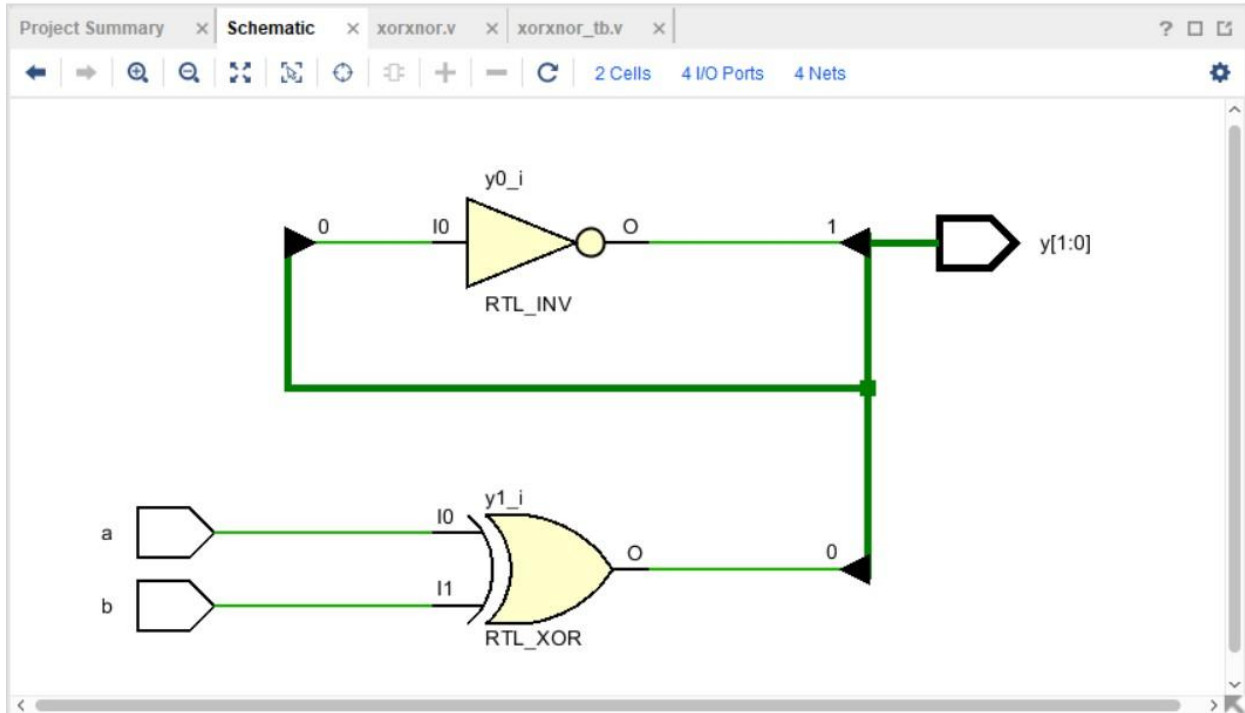


XOR and XNOR:



```
Project Summary x Schematic x xorxnor.v x xorxnor_tb.v x
C:/Users/P.SATHWIKA/universal/universal.srscs/sources_1/new/xorxnor.v

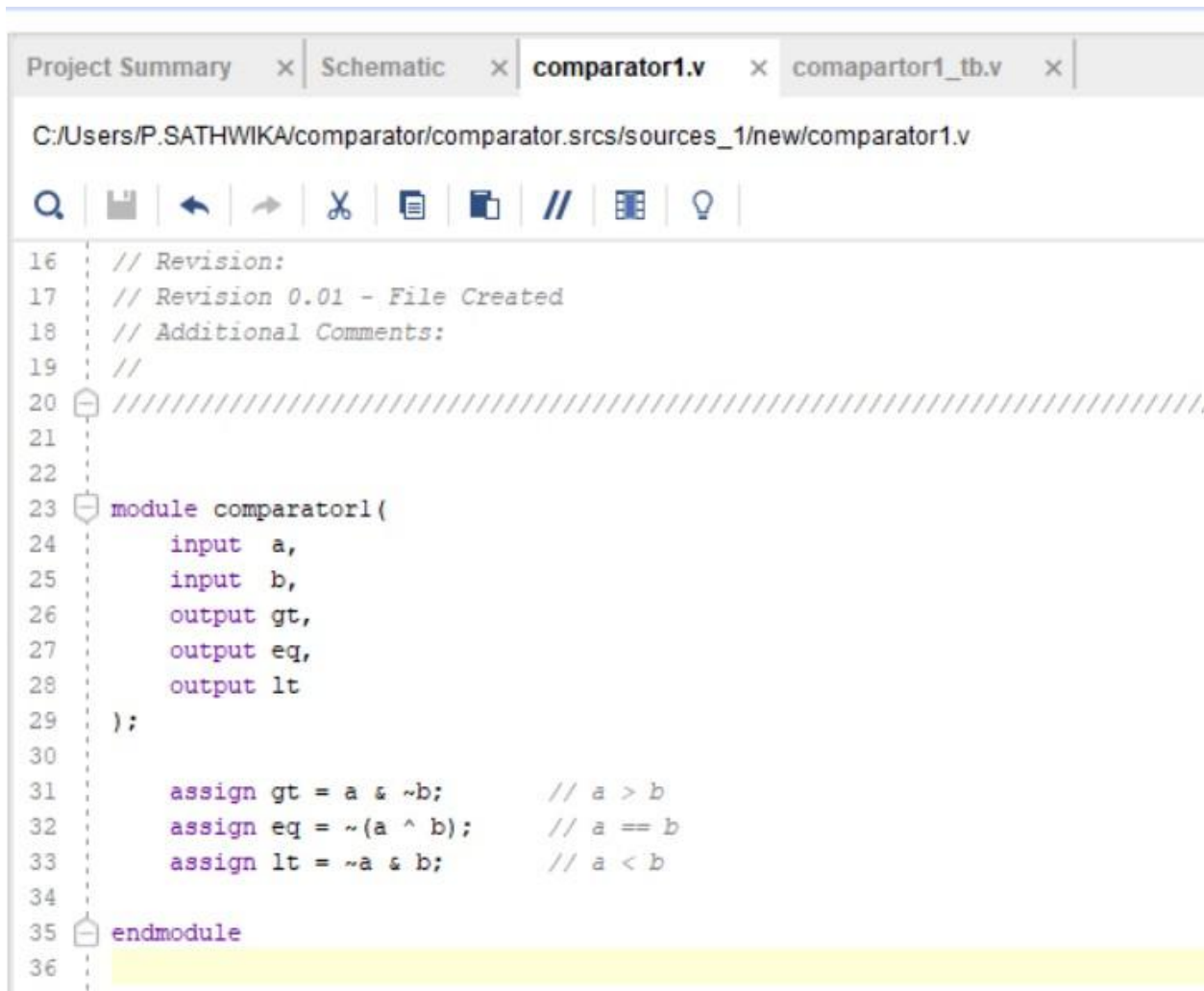
11 // Tool Versions:
12 // Description:
13 //
14 // Dependencies:
15 //
16 // Revision:
17 // Revision 0.01 - File Created
18 // Additional Comments:
19 //
20 ///////////////////////////////////////////////////////////////////
21
22
23 module xorxnor(
24     output [1:0] y,
25     input a,
26     input b
27 );
28     // y[1] gets XNOR, y[0] gets XOR explicitly through concatenation
29     assign y = { ~(a ^ b), (a ^ b) };
30 endmodule
31
```

```

23 `timescale 1ns / 1ps
24
25 module tb_xorxnor();
26     reg a;
27     reg b;
28     wire [1:0] y;
29
30     xorxnor uut (
31         .y(y),
32         .a(a),
33         .b(b)
34     );
35
36     initial begin
37         a = 0; b = 0;
38         #10;
39
40         a = 0; b = 1;
41         #10;
42
43         a = 1; b = 0;
44         #10;
45
46         a = 1; b = 1;
47         #10;
48         $finish;
49     end
50
51     initial begin
52         $monitor("Time = %0t ns | a = %b, b = %b | y (XNOR, XOR) = %b (y[1]=%b, y[0]=%b)",
53             $time, a, b, y, y[1], y[0]);
54     end
55
56 endmodule
57

```

1-BIT COMPARATOR:



The screenshot shows a Verilog code editor with the following tabs: Project Summary, Schematic, comparator1.v, and comapartor1_tb.v. The file path is C:/Users/P.SATHWIKI/comparator/comparator.srcs/sources_1/new/comparator1.v. The code defines a module named comparator1 with inputs a and b, and outputs gt, eq, and lt. The logic is implemented using assign statements: gt = a & ~b; eq = ~(a ^ b); and lt = ~a & b. The code is highlighted in yellow.

```
16 // Revision:
17 // Revision 0.01 - File Created
18 // Additional Comments:
19 //
20 ///////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////
21
22
23 module comparator1(
24     input a,
25     input b,
26     output gt,
27     output eq,
28     output lt
29 );
30
31     assign gt = a & ~b;           // a > b
32     assign eq = ~(a ^ b);       // a == b
33     assign lt = ~a & b;         // a < b
34
35 endmodule
36
```

```

21
22
23 module compartori_tb;
24     reg a;
25     reg b;
26     wire gt;
27     wire eq;
28     wire lt;
29
30     comparator_lbit uut (
31         .a(a),
32         .b(b),
33         .gt(gt),
34         .eq(eq),
35         .lt(lt)
36     );
37
38     initial begin
39
40         a = 0; b = 0;
41         #10;
42         a = 0; b = 1;
43         #10;
44         a = 1; b = 0;
45         #10;
46         a = 1; b = 1;
47         #10;
48
49         $finish;
50
51     end
52
53     initial begin
54         $monitor("Time=%0t | a=%b b=%b | a>%b a=%b a<%b",
55             $time, a, b, gt, eq, lt);
56     end
57

```

